

Lab 08 Inheritance and Polymorphism Cont..

Objective:

The purpose of this Lab is to practically familiarize students with the concept of Inheritance and Polymorphism.

Student will learn:

- To define a subclass from a superclass through inheritance.
- To override instance methods in the subclass.
- To discover polymorphism and dynamic binding.
- Promoting software reusability through inheritance.
- Casting and explain why explicit down casting is necessary.

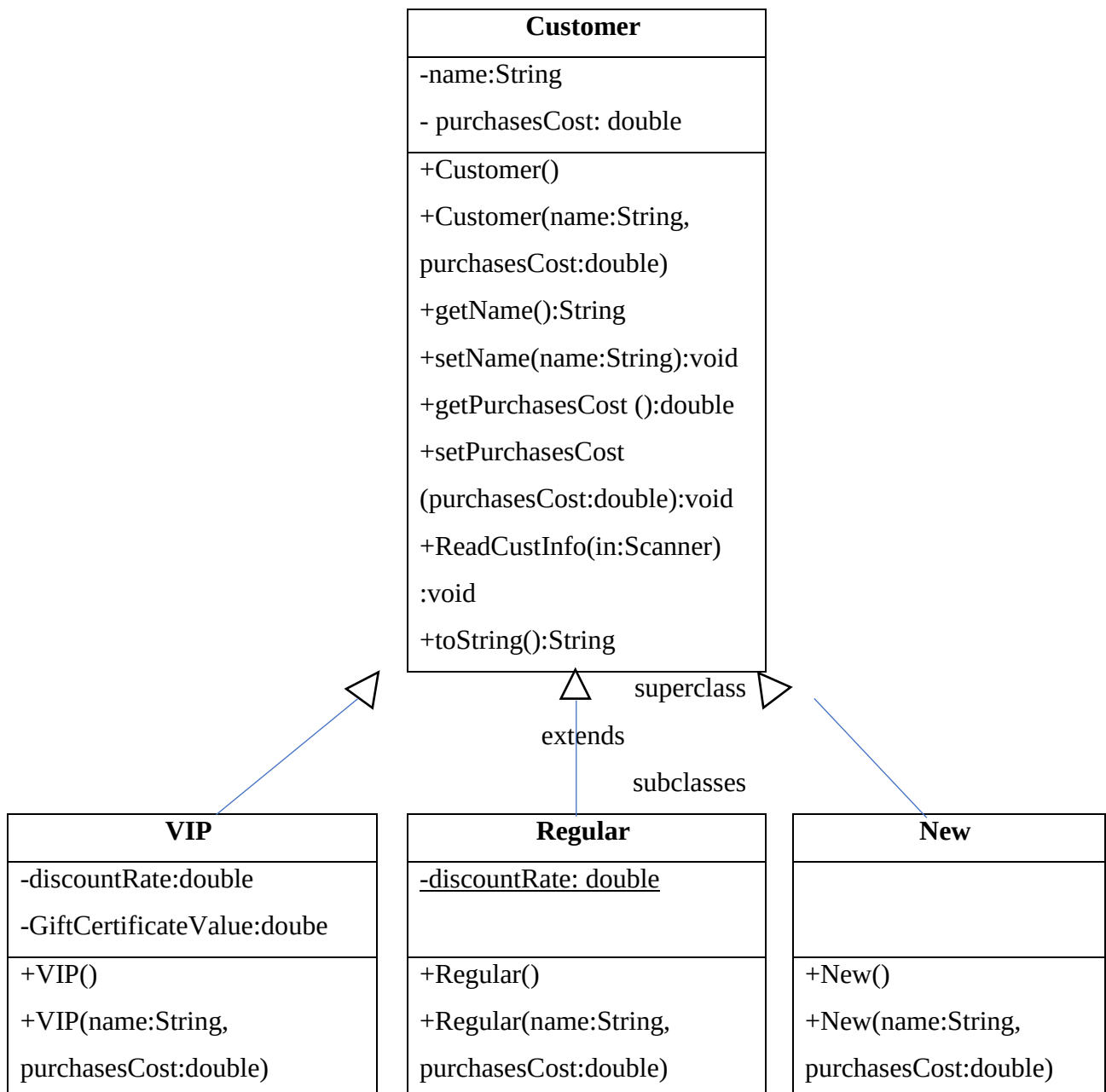
Activity Description:

You are asked to write a program for a shopping store that calculates the total payments for different customers regular, new, or VIP. The store offers **a special discount** only for their regular and VIP customers which deduct a defined percentage from the purchases cost. The store will offer a 100 SR. **Gift Certificate** for their VIP customers only when their purchased cost exceed 1000 .The store also keeps a record of all the customers and their total income. The following UML (next page,) describe the data field and methods of each class in this program.

Important Note:

- Initialize the discount rate which is 10%for VIP customers and 5% for regular customers.
- Initialize the GiftCertificateValue with zero for VIP customers.
- Initialize the name with empty string and the purchasesCost with zero
- Override "getPurchasesCost" in VIP and Regular classes to get the cost after discount.
- Override "toString" in VIP and Regular classes to add the cost after discount to what already displayed in the super class.
- In the main class:
 - Ask the user to enter number of the customers for the current day.
 - Create an array of customer (declared type)

- Start to fill the array by asking the user about the customer name, Purchases' Cost and type (V,R,N) and apply the polymorphism concept to decide the type of customers: VIP, Regular, New (actual type).
- Print as in the sample output:
 - customers' information with the customer type (**Hint:** use instanceof)
 - check and show if the VIP customer won a gift certificate. (**Hint:** Call GiftCertificate method in VIP class)
 - Print total income for that day



+getPurchasesCost():double
+ GiftCertificate():doub
+toString():String

+getPurchasesCost():double
+toString():String



Sample Output:

Enter number of customers for the current day:3

Enter customer's name:Yara

Enter customer's Purchases Cost:250

Enter customer's type (V for VIP, R for Regular and N for New):n

Enter customer's name:Lena

Enter customer's Purchases Cost:300

Enter customer's type (V for VIP, R for Regular and N for New):R

Enter customer's name:Lulu

Enter customer's Purchases Cost:1200

Enter customer's type (V for VIP, R for Regular and N for New):v

New Customer name:Yara , his/her purchases' Cost:250.0

Regular Customer name:Lena , his/her purchases' Cost:300.0 , cost after discount= 285.0

Congratulation for the next customer. You Won 100.0 SR. gift certificate

VIP Customer name:Lulu , his/her purchases' Cost:1200.0 , cost after discount= 1080.0

Current day income = 1615.0SR.